PROJECT DOCUMENTATION

AI-POWERED CUSTOMER SUPPORT & SERVICE AUTOMATION

SALESFORCE IMPLEMENTATION

Centralized customer records • Case routing • AI-assisted summaries • Analytics



Prepared for academic demonstration and project evaluation

EXECUTIVE SUMMARY

The AI-Powered Customer Support & Service Automation project demonstrates how Salesforce can be configured to organize customer information, merchandise, orders, and support cases in one Lightning application. The implementation combines custom objects and lookup relationships with case-assignment flows, support queues, an AI-assisted summary field powered by Prompt Builder, and reporting components for operational visibility.

The central workflow begins with customer, product, and order records. A User Case references the related order and merchandise, and records case type, priority, and issue category. An administrator can use assignment automation and queues to support routing. The AI Summary field is connected to the activated MetaDataInspection prompt template; the summary-generation control was used successfully on a sample case. The Customer Support Case Analysis report and Customer Support Dashboard were also created and reviewed.

The document follows the major structure of the supplied EventForce Salesforce implementation reference, but all examples and design descriptions are adapted to the customer-support project. It separates implemented and observed features from items that were not confirmed or were intentionally deferred. In particular, the three-day reminder flow was skipped, and a formal approval process, Agentforce configuration, Apex code, and a Salesforce metadata export were not verified as part of this build.

Document Purpose

This report records the project objectives, user problems, requirements, data model, app structure, automation, AI summary setup, reporting, verification activities, maintenance guidance, limitations, and future scope. It is intended to support project evaluation and provide a practical handover reference for future administrators.

Implementation Snapshot

Area	Current documented state	Evidence / note
Lightning app	Configured and visible in App Launcher	Custom app navigation confirmed
Custom data model	Customer, Merchandise, Orders, User Case	Sample records created and linked
Case routing	Two relevant flows appear active in Flow Manager	End-to-end routing should be retested before production use
Queues	Damage Support Queue and Warranty Support Queue exist	Existence confirmed in Setup
AI Summary	Prompt template activated and linked to the field	Generation button used; summary generated on a sample case
Analytics	Customer Support Case Analysis report and Customer Support Dashboard	Created and reported working by project owner
Deployment metadata	Not retrieved into a Salesforce DX source project	CLI / VS Code retrieval was deferred
Deferred / unverified	Three-day reminder flow, approval process, validation rules, Agentforce, Apex	Do not represent as completed unless subsequently implemented

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Reading this report

Status terms are used carefully: “Configured / observed” means confirmed in the build history or visible in the supplied Salesforce screenshots; “Not verified” means the implementation was not confirmed; “Future scope” means it is not represented as complete.

1. INTRODUCTION

1.1 Project Overview

The AI-Powered Customer Support & Service Automation system is a Salesforce-based solution designed to make customer support data easier to manage and support common case-handling activities. A custom Lightning app provides navigation across Customer, Merchandise, Orders, User Cases, and Knowledge. The custom data model connects a customer to an order and connects an order and merchandise to a support case.

The implementation adds assignment-related automation, two support queues, and an AI Summary field configured through Salesforce Prompt Builder. A report and dashboard provide a starting point for reviewing case information. The project is a practical configuration-based demonstration of Salesforce CRM features rather than a separate web application or a source-code-heavy implementation.

1.2 Purpose

- Centralize customer, merchandise, order, and support-case information in Salesforce.
- Represent relationships between customer transactions, products, and the support issues raised about them.
- Support case routing through assignment flows and Damage / Warranty queues.
- Provide AI-assisted summaries to help a support user understand a case more quickly.
- Create report and dashboard views that improve visibility into the support workload.
- Document the configuration, known limitations, verification status, and future improvements.

1.3 Project Objectives

Objective	Expected outcome
Data organization	One consistent Salesforce record model for customers, merchandise, orders, and cases.
Relationship tracking	A support case can reference its related order and product.
Routing support	Assignment flows and queues provide a basis for routing cases by case details.
AI assistance	Users can invoke Prompt Builder to generate an AI Summary on a User Case.
Analytics	A saved report and dashboard make case data easier to inspect.
Maintainability	Clear documentation supports future testing, enhancement, and troubleshooting.

1.4 Scope

The scope covers custom objects and fields, lookup relationships, the Lightning app, sample records, case-assignment flows and queues, Prompt Builder integration, a case report, and a dashboard. A three-day reminder, approval process, broader security configuration, Agentforce, Apex, and metadata export are not claimed as completed.

2. IDEATION PHASE

2.1 Problem Statement

Customer-support work becomes difficult to coordinate when customer details, product information, orders, and case details are kept separately. Support staff may have to navigate between records, manually decide where a case belongs, and summarize issue details before responding. Managers also need a quick view of the types and priorities of cases being handled.

ID	User / stakeholder	Need	Problem / pain point	Desired outcome
PS-1	Support Agent	Review a case with its order and product context	Related information can be spread across multiple records	A single case record links to the relevant order and merchandise.
PS-2	Support Lead	Route cases to the appropriate team	Manual assignment can delay work or create inconsistent ownership	Assignment automation and support queues guide routing.
PS-3	Support Agent	Understand the issue quickly	Reading and summarizing case information takes time	AI-assisted summary generation from a configured prompt.
PS-4	Service Manager	Review case priorities and patterns	Raw record lists provide limited operational visibility	Reports and dashboards group case information for review.
PS-5	Salesforce Administrator	Keep the app maintainable	Flows, prompt configuration, and relationships need monitoring	Structured documentation and repeatable checks.

2.2 User Persona: Support Agent

Persona aspect	Description
Role	Support Agent / Case Handler
Primary responsibility	Review user cases, inspect order and product context, update case records, and coordinate the next action.
Think and feel	Wants to respond accurately and avoid spending unnecessary time searching for context.
Hear	Receives customer concerns, hand-off notes, and internal routing instructions.
See	Case priority, case type, issue category, linked order, linked merchandise, and AI Summary.
Say and do	Opens a case, checks its related records, reviews the AI-generated summary, and follows the assigned workflow.
Pain points	Fragmented context, repetitive summarization, unclear routing, and limited workload visibility.
Gains	Linked records, a generated case summary, routing support, and consolidated analytics.

2.3 Brainstorming and Idea Prioritization

Option	Approach	Assessment
Manual handling	Keep customer, order, and case notes separately and assign cases manually.	Quick to begin but repetitive and difficult to standardize.
Spreadsheet tracking	Maintain support cases and routing information in sheets.	Limited record relationships, access control, and workflow integration.
Standalone support application	Build a separate ticketing UI and backend.	More development and hosting effort than required for a configuration-focused project.
Salesforce CRM	Use custom objects, lookup fields, flows, Prompt Builder, and analytics.	Selected because it supports structured data and configurable automation in one platform.

Decision: Salesforce was chosen as the implementation platform. The project emphasizes declarative configuration so the records, routing setup, AI summary, and analytics are visible in a single Lightning application.

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer / Case Handling Journey

Stage	Activity	Salesforce support
1. Customer context	Create or locate a customer record.	Customer object stores name, email, phone, and customer type.
2. Product context	Create or locate merchandise.	Merchandise object stores product code, category, warranty, and active status.
3. Order context	Create an order and associate the customer and product.	Orders object has lookup fields to Customer and Merchandise.
4. Case intake	Create a User Case and select case type, priority, issue category, order, and product.	User Case object stores case details and lookup relationships.
5. Assignment and handling	Review assignment automation and queue ownership.	Two assignment-related flows appear active in Flow Manager; queue existence was confirmed.
6. AI assistance	Invoke the AI Summary control and review the response.	MetaDataInspection prompt template is linked to AI Summary; sample generation was completed.
7. Analysis	Review a report and a dashboard.	Customer Support Case Analysis and Customer Support Dashboard were created and reviewed.

3.2 Functional Requirements

ID	Requirement	Description	Status
FR-01	Customer management	Store customer name, email, phone, and customer type.	Configured / observed
FR-02	Merchandise management	Store product code, category, warranty period, and active flag.	Configured / observed
FR-03	Order management	Create orders with an auto-number, date, status, customer, and product.	Configured / observed
FR-04	Case management	Capture case type, priority, issue category, order, and product.	Configured / observed
FR-05	Case assignment	Support assignment logic through Salesforce flows.	Flows listed active; end-to-end routing needs retest
FR-06	Support queues	Provide Damage Support Queue and Warranty Support Queue.	Queues exist
FR-07	AI case summary	Generate a summary from the configured prompt template.	Generation tested on a sample case
FR-08	Case analytics	Provide a case-analysis report and dashboard chart.	Created and reviewed
FR-09	Validation rules	Prevent invalid or incomplete records based on business rules.	Not verified in this build
FR-10	Approval process	Control defined case or service decisions through approval.	Not verified / deferred
FR-11	Three-day reminder	Send a reminder after a configured period.	Not built; skipped
FR-12	Agentforce support	Offer an AI agent experience where licensed and configured.	Not verified / future scope
FR-13	Deployable source	Export metadata for version control and deployment.	Not completed

3.3 Non-Functional Requirements

Attribute	Requirement / design consideration	Current note
Usability	Use a clear Lightning navigation structure and page layout.	Custom app and User Case page configured.

Attribute	Requirement / design consideration	Current note
Security	Limit access according to user profile, permission, and sharing needs.	App assigned to System Administrator and Standard User profiles; broader security model not verified.
Reliability	Flows and AI summary generation should produce predictable results.	AI generation observed; flow end-to-end tests remain advisable.
Performance	Records and reports should remain responsive for the expected sample workload.	No load or scale test was performed.
Maintainability	Use structured fields, named flows, and documented processes.	Documentation prepared; metadata export remains future work.
Data integrity	Use appropriate field types and valid lookup references.	Lookup relationships used; explicit validation rules not confirmed.

3.4 High-Level Data Flow

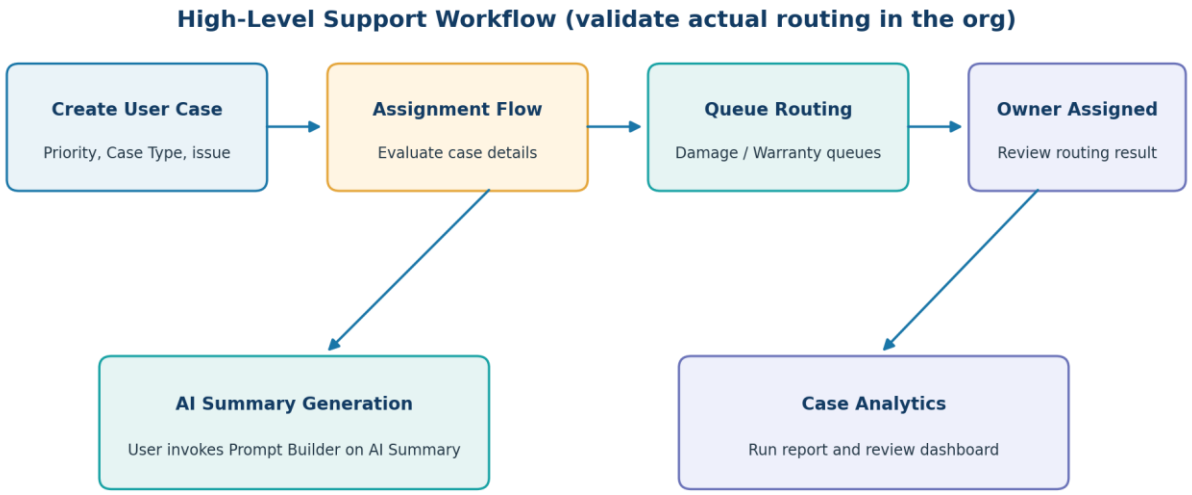


Figure 1. High-level case-handling flow. The diagram shows the intended path; routing outcomes should be verified with test records.

3.5 Technology Stack

Layer	Technology / component	Role
Platform	Salesforce Lightning Experience	Cloud CRM and configuration environment.
User interface	Lightning app and record pages	Navigation and presentation of records.
Data layer	Salesforce custom objects and fields	Customer, merchandise, order, and case records.
Relationships	Lookup fields	Connect orders to customers and products, and cases to orders and products.
Automation	Salesforce Flow Builder	Case-assignment-related logic.
AI assistance	Salesforce Prompt Builder / Einstein Generative AI field integration	Generate AI Summary on a User Case.
Analytics	Salesforce Reports and Dashboards	Case analysis and visualization.
Documentation	GitHub Markdown repository	Project documentation and diagrams; no source metadata export yet.

3.6 Assumptions and Constraints

Item	Documented assumption / constraint
Platform access	Configuration was performed in a Salesforce Lightning org with access to Setup and Prompt Builder.
Licensing	Generative AI availability depends on org settings, permissions, and licensing; the successful test applies to the observed org.
Sample data	Testing used sample customer, merchandise, order, and User Case records.
Deployment	A metadata package has not been retrieved or deployed into another org.
Deferred requirements	Three-day reminders, formal approval, validation rules, Agentforce, and Apex logic were not confirmed in this implementation.

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

Problem	Configured solution	Expected benefit
Customer, order, product, and case information needs structure.	Separate Customer, Merchandise, Orders, and User Case objects with lookup relationships.	Related context can be displayed without duplicating all information.
Case routing can require manual triage.	Assignment-related flows and named Damage / Warranty queues.	Provides a basis for consistent case routing; confirm through live tests.
Support staff need an overview of an issue.	Prompt Builder template connected to the AI Summary field.	Reduces manual effort when generating a first-draft summary.
Managers need visibility into support cases.	Customer Support Case Analysis report and Customer Support Dashboard.	Makes case information easier to review.
Administration needs clear operational knowledge.	Project documentation and a data model diagram.	Supports handover and future troubleshooting.

4.2 Proposed Solution

The proposed solution is a Lightning application with a small data model for support operations. Users create records in the order that business context is known: customer, merchandise, order, then User Case. The case form references the order and product. Flows and queues support routing. Prompt Builder provides an optional AI-generated summary, while a report and dashboard support basic case analysis.

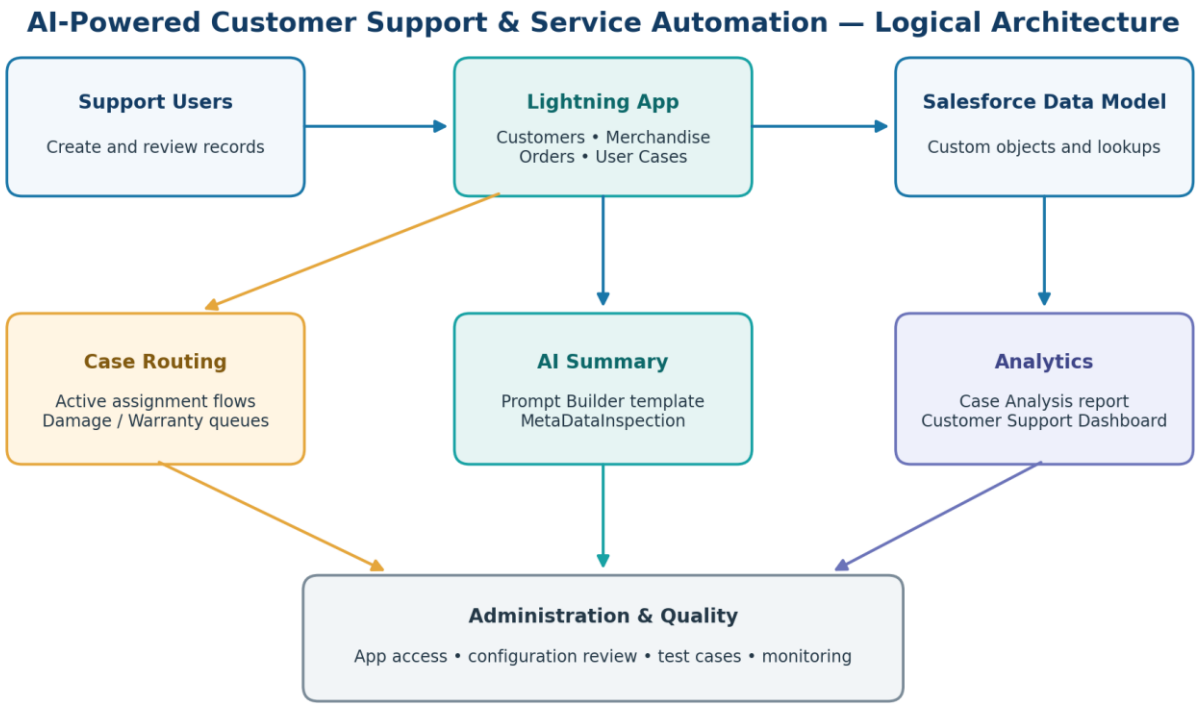


Figure 2. Logical solution architecture for the current Salesforce implementation.

4.3 Data Model and Relationships

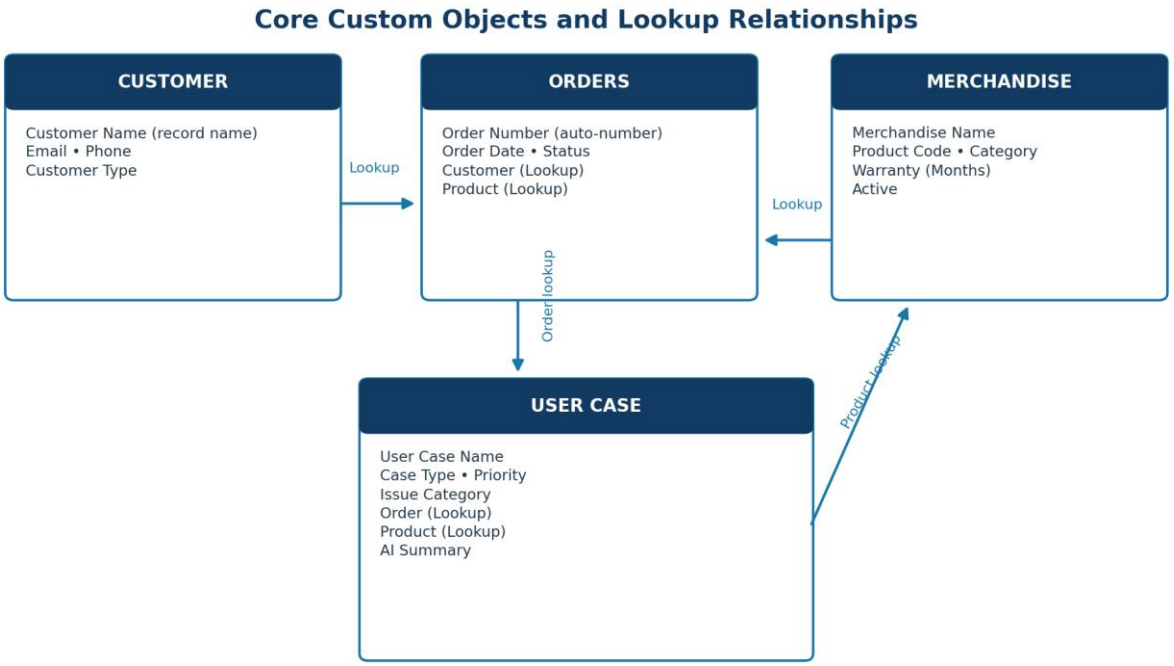


Figure 3. Core Salesforce custom objects and lookup relationships. Field names are simplified for readability.

Relationship design: Orders uses lookup fields to Customer and Merchandise. User Case uses lookup fields to Orders and Merchandise. This links the support issue to both the transaction and the affected product. The exact API names should be read from Object Manager or an exported metadata package before creating a deployment manifest.

4.4 Data Dictionary

Object	Field	Type / notes
Customer	Customer Name	Record name, text
Customer	Email	Email
Customer	Phone	Phone
Customer	Customer Type	Picklist: Individual, Business
Merchandise	Merchandise Name	Record name, text
Merchandise	Product Code	Text
Merchandise	Category	Picklist; values as configured in the org
Merchandise	Warranty (Months)	Number
Merchandise	Active	Checkbox
Orders	Order Number	Auto Number; format ORD-{0000}, starting at 1
Orders	Order Date	Date
Orders	Status	Picklist: New, Shipped, Delivered
Orders	Customer	Lookup to Customer
Orders	Product	Lookup to Merchandise
User Case	User Case Name	Record name
User Case	Case Type	Picklist: Damage, Warranty, Delivery
User Case	Priority	Picklist: Low, Medium, High
User Case	Issue Category	Picklist: Delivery, Damage, Warranty
User Case	Product	Lookup to Merchandise
User Case	Order	Lookup to Orders
User Case	AI Summary	Long Text Area; field API name shown as AI_Summary__c

Relationship and Data Quality Notes

- Lookup fields connect records; they do not automatically enforce every business rule required for case handling.
- Order Status is documented with New, Shipped, and Delivered values in the current configuration.
- User Case Case Type and Issue Category both include Damage, Warranty, and Delivery; their business meaning should be kept consistent.
- Before deployment, verify field API names, requiredness, picklist restrictions, record type effects, and field-level access in Object Manager.

5. PROJECT PLANNING AND SCHEDULING

5.1 Planning Approach

The project was implemented iteratively through Salesforce Setup and Lightning UI. The sequence prioritized the data model, navigation, sample records, case automation and queues, AI Summary configuration, and analytics. The work was guided by visible requirements and screenshots, and the implementation was checked incrementally rather than deploying a complete source package. No sprint duration, story-point estimates, team allocation, or release dates were supplied; this roadmap documents implementation order without inventing a schedule.

5.2 Implementation Roadmap

Phase	Activities	Status in this project
1. Requirements	Identify the customer-support problem, objects, and core workflow.	Completed for the documented scope.
2. Data setup	Create custom objects and fields; establish lookups.	Configured and sample records created.
3. User interface	Create tabs and Lightning app; configure User Case page.	Configured; app visible in App Launcher.
4. Routing	Create assignment flows and support queues.	Flows appear active and queues exist; routing should be regression-tested.
5. AI assistance	Enable generative AI, create and activate prompt template, assign it to field.	Configured; sample generation worked.
6. Analytics	Create case report and dashboard.	Created and opened successfully per user verification.
7. Final quality checks	Test edge cases, permissions, and deployment readiness.	Partially completed; see testing and limitations.
8. Source control	Retrieve Salesforce metadata and add it to GitHub.	Deferred; repository currently documents the project.

5.3 Milestones

Milestone	Deliverable	Completion evidence
M1 — Data model	Customer, Merchandise, Orders, User Case	Objects, fields, and sample records.
M2 — App navigation	Lightning app and object tabs	App searchable; five navigation items visible.
M3 — Routing foundations	Case assignment flows and queues	Flows shown active; queues found in Setup.
M4 — AI field	Prompt template linked to AI Summary	Template linked; generation button used.
M5 — Analytics	Case report and support dashboard	Report and dashboard opened successfully.
M6 — Submission	Demo link and GitHub documentation	Docs repository created; verify demo sharing.

6. PROJECT DEVELOPMENT AND CONFIGURATION

6.1 Salesforce App and Navigation

A Lightning app named “AI-Powered Customer Support & Service Automation” was created in App Manager. The app was configured to be available to the System Administrator and Standard User profiles. Its navigation includes Customers, Merchandise, Orders, User Cases, and Knowledge. After profile assignment was saved, the app appeared in the App Launcher.

The User Case record page was opened in Lightning App Builder. The AI Summary field was selected and connected to the activated Prompt Builder template. The screenshot evidence below records the selected template and field placement.

6.2 Custom Objects and Sample Records

The primary records used in the walkthrough include a sample customer, sample merchandise, an order linked to that customer and product, and a User Case linked to the order and product. This creates a connected example that can be followed during the demonstration.

- Customer: stores name, email, phone, and customer type.
- Merchandise: stores name, product code, category, warranty period, and active flag.
- Orders: automatically generates an order number and stores order date, status, customer lookup, and product lookup.
- User Case: stores case type, priority, issue category, product lookup, order lookup, and AI Summary.

6.3 Case-Assignment Flows

The Flow Manager list showed “Case Auto Assignment & SLA Flow” and “Auto Assign Case to Queue”. Their Active indicators were selected in the captured screen. The design intent is to use case details to support assignment and queue routing. Because a list status does not prove that every path works in real transactions, an end-to-end test should create a new test case for each routing scenario and inspect the resulting owner.

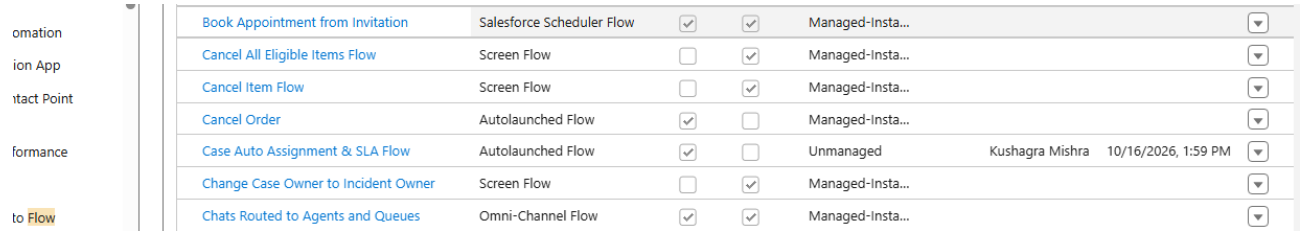


Figure 4. Salesforce Flow Manager screenshot showing the Case Auto Assignment & SLA Flow and Auto Assign Case to Queue entries. “Unmanaged” is a package/management classification, not a failure state.

6.4 Support Queues

Queue name	Purpose in the documented design	Verification status
Damage Support Queue	Queue intended to receive damage-related support cases.	Existence confirmed in Salesforce Setup.
Warranty Support Queue	Queue intended to receive warranty-related support cases.	Existence confirmed in Salesforce Setup.

Queue membership and actual owner updates should be validated with test users and test User Case records. Ensure the relevant support agents are members of the queues and have sufficient object permissions before operational use.

6.5 AI Summary with Prompt Builder

The prompt template named “MetaDataInspection” was activated. On the User Case Lightning record page, the AI Summary field (AI_Summary__c) was selected and the prompt template was set in the Einstein Generative AI section. The template was invoked from a sample User Case and a summary was produced. However, the source prompt wording supplied during setup is inspection-oriented (for example, it refers to inspection status, date, and observations) rather than explicitly to customer-support case details. Before using the field operationally, review and align the prompt instructions and input resources with the actual User Case fields.

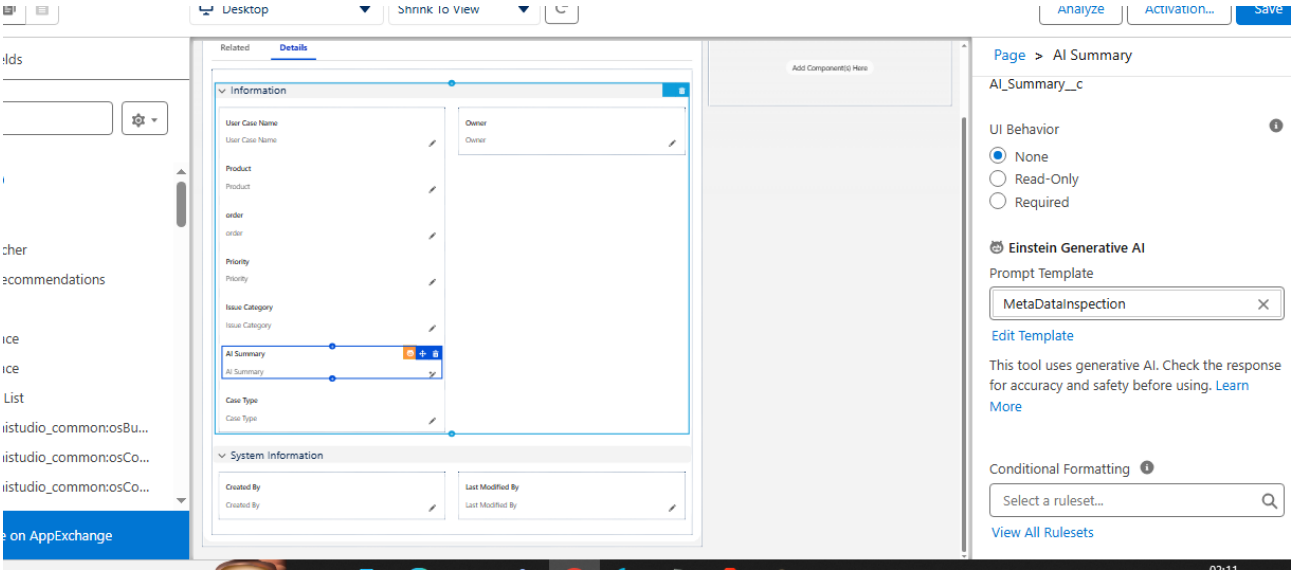


Figure 5. Lightning App Builder shows the AI Summary field with MetaDataInspection selected as its Prompt Template.

Responsible AI handling

AI-generated summaries are drafts. Support users should review them for accuracy, relevance, and sensitive content before using them to make decisions or communicate with customers.

6.6 Reports and Dashboards

A User Cases report named “Customer Support Case Analysis” was created and saved. The report builder was used to add case-related columns and a priority grouping. A dashboard named “Customer Support Dashboard” was created and a chart titled “Cases by Priority” was added from the report. The user confirmed that the report and dashboard were working.

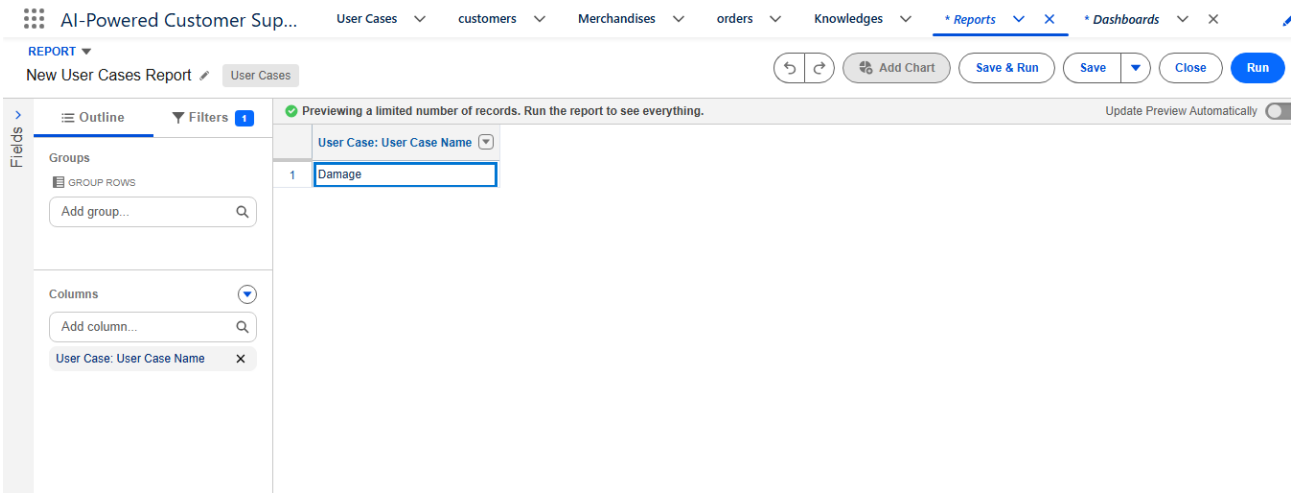


Figure 6. Salesforce Report Builder with the User Cases report and a sample case row. The report shown here was later named Customer Support Case Analysis.

6.7 Security and Access Considerations

The app's User Profiles section showed System Administrator and Standard User in the selected profiles list. This confirms app availability for those profiles, but it does not by itself confirm object-level permissions, field-level security, record sharing, role hierarchy, or organization-wide defaults. Those settings should be reviewed separately if the app will be assessed for security completeness or used by real support teams.

6.8 Configuration Readiness Notes

Readiness item	Current state / recommended check
App discoverability	Confirmed visible in App Launcher after profile assignments were saved.
Record page	AI Summary field is on the User Case page and Prompt Template is selected.
Flow ownership	Confirm each test case is assigned to the expected queue or user; active status alone is insufficient.
User access	Repeat tests as a Standard User and verify access to all linked records, prompts, reports, and dashboards.
Portability	Retrieve metadata before attempting repeatable deployment or source-controlled releases.

7. FUNCTIONAL AND PERFORMANCE TESTING

7.1 Test Strategy

Testing focused on whether the principal UI components could be opened, whether sample records were connected, whether AI summary generation could be invoked, and whether the saved report and dashboard could be displayed. Tests that were not actually performed are marked accordingly.

7.2 Functional Test Cases

Test ID	Scenario	Expected / observed result	Status
TC-01	Open custom Lightning app	App is found through App Launcher.	Pass — observed
TC-02	Navigate core object tabs	Customers, Merchandise, Orders, and User Cases are available.	Pass — confirmed visible
TC-03	Open sample Customer record	Customer record displays profile fields.	Pass — sample record created
TC-04	Open sample Merchandise record	Product details are stored and accessible.	Pass — sample record created
TC-05	Link an Order	Order references the customer and merchandise.	Pass — linked sample record created
TC-06	Create a User Case	Case contains case type, priority, issue category, order, and product.	Pass — sample record created
TC-07	Generate AI Summary	Prompt template can be invoked from the AI Summary field.	Pass — summary generated on sample
TC-08	Check flow status	Assignment flows are listed as active in Flow Manager.	Pass — status observed; not a full routing test
TC-09	Confirm support queues	Damage and Warranty queues are listed in Setup.	Pass — existence confirmed
TC-10	Open case report	Customer Support Case Analysis displays case data.	Pass — user confirmed working
TC-11	Open dashboard	Customer Support Dashboard displays a chart.	Pass — user confirmed working
TC-12	Validation-rule edge cases	Invalid cases should be prevented.	Not tested / validation rules not confirmed
TC-13	Approval workflow	A request should be submitted and approved/rejected.	Not verified
TC-14	Three-day reminder	Reminder action should run after the defined wait.	Not implemented; skipped
TC-15	Queue owner assignment	New cases route to the intended queue for each case type.	Needs end-to-end regression test
TC-16	Deployment to another org	Retrieve, deploy, and validate metadata in a target org.	Not performed

7.3 Performance and Reliability

No controlled load test, large-volume test, or production performance measurement was performed. The current evaluation is functional and UI-based using sample records. Before broader use, test flows with multiple record combinations, confirm queue membership and owner assignment, check prompt responses across different case types, and ensure that report totals are correct when more records are present.

7.4 Acceptance Criteria

- The Lightning app is discoverable to the intended users.
- Core objects and lookup relationships are usable in sample records.

- The AI Summary generation action produces a reviewable response.
- The report and dashboard open and show relevant case information.
- Unverified and deferred requirements are documented transparently.

7.5 Evidence and Retest Priority

Priority	Retest activity	Why it matters
1	Create a fresh Damage case and confirm owner assignment.	Validates the key routing decision in a real transaction.
2	Repeat routing tests for Warranty and Delivery.	Checks each case-type branch and queue mapping.
3	Generate summaries from different case records and review accuracy.	Checks consistency and reduces reliance on a single sample.
4	Reconcile report totals with User Case list views.	Confirms the report filters and dashboard source data are correct.
5	Test as Standard User.	Confirms practical access and least-privilege behavior.

8. MAINTENANCE, MONITORING, AND TROUBLESHOOTING

8.1 Maintenance Activities

Maintenance keeps the application dependable as case volumes and business needs change. An administrator should periodically review assignment flows, queue configuration, prompt templates, record page assignments, field definitions, report filters, and dashboard source reports. Changes should be validated in a safe environment before being applied to a shared or production org.

Area	Routine check	If an issue is found
Flows	Check active status, versions, error emails, and failed interviews.	Review the fault path, entry criteria, field references, and record updates; test after correction.
Queues	Verify queue names, members, and object support.	Correct queue membership or routing configuration and retest owner assignment.
Prompt Builder	Confirm template activation, input resources, and field assignment.	Verify required resource fields, permission/licensing, and regenerate using a test record.
AI Summary field	Confirm the generation action remains visible on the record page.	Review Lightning page activation, selected template, and component configuration.
Reports	Check filters, groupings, available fields, and record counts.	Run the report and compare it with User Case list data.
Dashboard	Check the source report, chart title, refresh, and access.	Open the report independently and verify dashboard folder sharing and source report.
Security	Review profile, permission set, sharing, and field-level access.	Re-test as a standard support user; grant only required permissions.
Data quality	Check required values and link accuracy.	Define and validate field rules before enforcing them in live data.

8.2 Monitoring Indicators

- Assignment accuracy: percentage of test cases owned by the expected queue or user.
- Flow health: failed flow interviews, fault emails, and unexpected owner updates.
- AI response quality: summaries should reflect case information without invented details.
- Report consistency: dashboard totals should reconcile with report results and list views.
- Access health: intended users can access required records and cannot access restricted information.

8.3 Troubleshooting Workflow

1. Reproduce the issue using a dedicated test record and note the exact error or unexpected value.
2. Identify the affected component: data, lookup relationship, flow, queue, prompt template, record page, report, or access setting.
3. Inspect recent configuration changes and execution details before changing the design.
4. Apply the smallest correction possible in a safe environment.
5. Re-run the original test and relevant regression tests; document the outcome.
6. Only then promote the change to the shared org or deployment target.

9. ADVANTAGES AND LIMITATIONS

9.1 Advantages

Centralized record context: Customer, merchandise, order, and case data are organized using custom objects and lookup relationships.

Improved case handling: Assignment-related flows and named support queues provide a foundation for routing by case details.

AI-assisted productivity: Prompt Builder can generate an initial case summary that a support user can review.

Operational visibility: The case report and dashboard provide a starting point for examining case priorities.

Configurable platform: Salesforce Setup, Flow Builder, Lightning App Builder, and Prompt Builder support future changes.

Handover documentation: The report records object fields, relationships, configuration, test evidence, and known gaps.

9.2 Limitations

Limitation	Implication
No metadata export to source control	The GitHub repository currently documents the project; it is not yet a deployable Salesforce DX project.
Routing not fully regression-tested	Active status alone does not prove that every decision path assigns the correct owner.
Approval process not verified	A formal approval workflow should not be claimed as delivered until tested.
Three-day reminder not implemented	The demo guideline names this flow, but it was deliberately skipped.
Validation rules and detailed sharing not verified	Record validation and role-based access need separate configuration and tests if required.
Agentforce and Apex not confirmed	The implementation documented here relies on the observed configuration and does not claim an Agentforce agent or Apex logic.
Sample-data evaluation only	The report and dashboard have not been load-tested against a large support workload.

10. CONCLUSION

The AI-Powered Customer Support & Service Automation project demonstrates how Salesforce can be used to build a compact, structured support workflow. Customer, Merchandise, Orders, and User Case records are modeled as related objects inside a custom Lightning app. Assignment-related flows and support queues provide routing foundations; Prompt Builder is connected to the AI Summary field and was used to generate a sample response; a case analysis report and dashboard provide initial analytics.

The implementation provides hands-on experience with Salesforce objects, fields, lookup relationships, Lightning navigation, Flow Manager, queues, Prompt Builder, and reporting. Its current value is as a functioning academic demonstration. Before treating it as production-ready, the routing logic should be tested end to end, access permissions and validation should be reviewed, and the unimplemented reminder/approval features should either be completed or kept explicitly out of the delivered scope.

A key follow-up is metadata retrieval and version control. The GitHub repository contains project documentation, but the Salesforce configuration has not yet been retrieved as deployable source. This does not change the working org, but it limits repeatable deployment and recovery until addressed.

11. FUTURE SCOPE

Priority	Enhancement	Benefit / next action
High	Export deployable Salesforce metadata	Use Salesforce CLI / VS Code or another supported metadata retrieval workflow to capture objects, fields, flows, tabs, app, reports, dashboards, and prompt metadata where supported; validate in a target org.
High	End-to-end routing tests	Create new test cases for Damage, Warranty, and Delivery; confirm which flow handles each scenario and inspect record owner changes.
High	Formal security and validation review	Review field-level security, object permissions, record sharing, validation rules, queue membership, and access as a Standard User.
Medium	Three-day reminder flow	Implement only after defining the date trigger, reminder channel, recipients, exclusions, and duplicate-notification behavior.
Medium	Approval process	Define entry criteria, approver, approval/rejection actions, and status updates before implementation.
Medium	Expanded support analytics	Add trends by case type, issue category, age, owner, and resolution status; verify filters and report types.
Medium	Agentforce support experience	Evaluate licensing, permissions, grounding, supported actions, and safety requirements before configuration.
Low	Richer case knowledge base	Populate Knowledge articles and link relevant help content to case types and products.
Ongoing	Continuous monitoring	Track flow faults, prompt quality, case routing accuracy, report integrity, and user feedback.

12. APPENDIX

Appendix A — Demo Video Outline

Sequence	What to show	Suggested duration
1. Introduction	Show the app title and explain the project purpose using on-screen text if recording silently.	10–15 sec
2. App navigation	Show Customers, Merchandise, Orders, User Cases, and Knowledge tabs.	15–20 sec
3. Sample data	Open a customer, merchandise item, order, and User Case; show lookup connections.	45–60 sec
4. AI Summary	Open User Case and show the generated AI Summary.	25–35 sec
5. Flows and queues	Show relevant flows in Setup and the Damage / Warranty queues; avoid claiming untested behavior.	30–45 sec
6. Analytics	Open Customer Support Case Analysis and Customer Support Dashboard.	30–45 sec
7. Conclusion	Return to the custom app home view.	5–10 sec

Appendix B — Configuration Inventory

Component	Name / details	Documentation status
Lightning App	AI-Powered Customer Support & Service Automation	Configured / app visible
Navigation	Customers, Merchandise, Orders, User Cases, Knowledge	Visible in app
Custom Object	Customer	Fields documented
Custom Object	Merchandise	Fields documented
Custom Object	Orders	Fields and lookups documented
Custom Object	User Case	Fields and lookups documented
Flow	Case Auto Assignment & SLA Flow	Shown active in Flow Manager; regression test recommended
Flow	Auto Assign Case to Queue	Shown active in Flow Manager; regression test recommended
Queue	Damage Support Queue	Existence confirmed
Queue	Warranty Support Queue	Existence confirmed
Prompt Template	MetaDataInspection	Activated, assigned to AI Summary, sample generation performed
Report	Customer Support Case Analysis	Created and user-confirmed working
Dashboard	Customer Support Dashboard	Created and user-confirmed working
Validation rules / approval / reminder / Agentforce	Not confirmed or not implemented in this scope	Do not describe as completed
Salesforce DX source export	Not performed	Future task

Appendix C — Glossary

Term	Meaning in this project
Custom object	A Salesforce data structure used to store a category of records.

Term	Meaning in this project
Lookup relationship	A field connecting one Salesforce record to another.
Flow	Salesforce declarative automation for evaluating conditions and performing actions.
Queue	A shared record owner or work queue that can be used for team-based handling.
Prompt Builder	Salesforce capability used here to configure a prompt template for AI-assisted text generation.
AI Summary	Long Text Area field on User Case, connected to the prompt template.
Report	A structured view of selected Salesforce records and fields.
Dashboard	A visual presentation of report-based metrics or charts.
Metadata	Configuration definitions that can be retrieved and versioned separately from ordinary business records.

Appendix D — Submission Checklist

- Fill the cover-page placeholders for student name and institution.
- Ensure the demo video is uploaded and its sharing access allows the evaluator to view it.
- Check that the GitHub repository is public if the submission requires public access.
- Remember that the current GitHub repository is documentation-only; deployable metadata export was deferred.
- Keep unimplemented features clearly labelled as future scope or not verified.

END OF PROJECT DOCUMENTATION